

Phase I Proposal

Topic AF254-0801: AI/ML-Generated Decoy Networks

1. Executive Summary

[Company Name] proposes to develop **DecoyGen-AI**, a software suite that leverages state-of-the-art machine learning and AI techniques to automatically generate, deploy, and manage highly realistic, dynamic decoy networks (“honey nets”) for Defensive Cyber Operations. DecoyGen-AI will mirror live network behaviors—users, infrastructure, data flows—while dynamically adapting to adversary tactics. Phase I will deliver a feasibility study evaluating candidate AI/ML approaches (e.g., generative adversarial networks, reinforcement learning, expert systems), define a modular software architecture, and demonstrate proof-of-concept decoy generation in a controlled sandbox environment.

2. Technical Objectives

1. **Survey & Evaluation:** Assess current AI/ML methods (GANs, RL, expert systems, blockchain-based ledgers) for decoy realism and adaptability.
 2. **Data Collection & Modeling:** Define data requirements and pipelines to ingest live network telemetry for training decoy behavior models.
 3. **Prototype Architecture:** Design a scalable, modular software framework supporting automated decoy creation, lifecycle management, and real-time adversary tracking.
 4. **Proof-of-Concept Demonstration:** Implement a minimal viable prototype that generates a simple decoy network, integrates basic ML-driven behavior, and logs attacker interactions stealthily in a sandbox.
-

3. Background & Innovation

Traditional honeypots suffer from static configurations and reveal themselves under expert scrutiny. Recent advances in generative modeling (e.g., GANs) and reinforcement learning enable creation of synthetic but realistic traffic patterns and dynamic adaptation. By combining these techniques with blockchain-backed state management and expert “white-hat” feedback loops, DecoyGen-AI will raise the bar against state-sponsored adversaries, providing:

- **Automation:** Fully automated decoy deployment and configuration with minimal operator input.
 - **Realism:** Decoys indistinguishable from production networks through continuous ML-driven adaptation.
 - **Insight:** Deep visibility into adversary TTPs without detection.
-

4. Phase I Approach

4.1 Methodology

- **Literature & Technology Review**
 - Analyze referenced works (Sun & Kim 2016; Dougherty 2020; Chong & Koh 2018) to derive success factors and limitations.
- **Algorithm Selection**
 - Evaluate GANs for generating synthetic traffic, RL for decoy behavior policies, expert systems for orchestrating scenario logic.
- **Data Pipeline Design**
 - Specify interfaces to ingest NetFlow, system logs, user activity metadata from live environments.
- **Architecture & Module Design**
 - Define core modules: Data Collector, Model Trainer, Decoy Orchestrator, Monitoring Dashboard, Blockchain Ledger.
- **Proof-of-Concept Implementation**
 - Build a sandbox that deploys virtual machines and network services via orchestration scripts, integrates a simple GAN-generated traffic emulator, and captures adversary activity stealthily.

4.2 Work Plan & Milestones

Task	Months 1–2	Months 3–4	Month 5	Month 6
Literature & Requirements	Review & selection			

Task	Months 1–2	Months 3–4	Month 5	Month 6
Data Pipeline & Modeling Plan		Design & prototyping		
Architecture Design			Module interfaces defined	
Prototype Development			Initial deployment & tests	
Demonstration & Reporting				Feasibility Report delivered

5. Key Personnel

Name	Role	Expertise
Dr. Alex Chen	Principal Investigator	10 years in AI/ML for cybersecurity; GAN & RL specialist
Ms. Priya Patel	Lead Software Architect	12 years in distributed systems & virtualization
Mr. Daniel Ortiz	Data Scientist	8 years in network telemetry analysis & anomaly detection
Mr. Robert Taylor	DevOps Engineer	7 years in cloud orchestration, containerization security

6. Company Overview

[Company Name] is a cyber-security innovator with a track record of delivering ML-driven defense solutions to DoD and commercial clients. We hold CMMC Level 3 certification and maintain an ITAR-compliant development environment. Our facilities include a secure lab sandbox with virtual network ranges and hardware-in-the-loop for testing advanced cyber defenses.

7. Facilities & Environment

- **Secure Lab:** Isolated testbed with VLAN-segmented networks, hardware firewalls, IDS sensors.
 - **Compute Resources:** GPU-accelerated servers for training deep learning models.
 - **Software Stack:** Kubernetes clusters, Terraform/Ansible for orchestration, blockchain testnet.
-

8. Phase I Deliverables

1. **Feasibility Study Report:** Evaluation of AI/ML methods, data requirements, high-level architecture.
 2. **Software Architecture Specification:** Detailed module designs, API definitions.
 3. **Proof-of-Concept Prototype:** Demonstrating automated decoy deployment and basic ML-driven adaptation in sandbox.
 4. **Phase II Plan:** Roadmap for scaling to TRL 6, including Risk Management Framework integration.
-

9. Conclusion

By the end of Phase I, [Company Name] will have identified the optimal AI/ML techniques and produced a validated software architecture for **DecoyGen-AI**, positioning us to deliver a fully functional prototype in Phase II. This solution promises to enhance Air Force and DoD cyber defense posture by diverting and analyzing advanced adversaries within realistic, self-evolving decoy environments.